

Exercice I.

[30 points]

Calculer les primitives suivantes:

1. $\int \ln |x| dx,$

3. $\int \sin^2 x \cos^2 x dx,$

2. $\int (\arcsin x)^2 dx,$

4. $\int e^x \sin x dx,$

Exercice II.

[10 points]

Calculer $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k}{n\sqrt{n^2 + k^2}}.$

Exercice III.

[20 points]

Soit $f : [0, 1] \mapsto \mathbb{R}$ une fonction continue.

1. Montrer que $\lim_{n \rightarrow +\infty} \int_0^{1 - \frac{1}{\sqrt{n}}} f(t^n) dt = f(0).$

2. Déduire la valeur de $\lim_{n \rightarrow +\infty} \int_0^1 f(t^n) dt.$

Exercice IV.

[40 points]

1. Montrer que les intégrales impropres $I = \int_1^{+\infty} \frac{\cos t}{t} dt$ et $J = \int_1^{+\infty} \frac{\cos^2 t}{t^2} dt$ sont convergentes.

2. Déduire la nature de $K = \int_1^{+\infty} \ln \left(1 + \frac{\cos t}{t}\right) dt.$

3. Montrer que l'intégrale impropre suivante $L = \int_1^{+\infty} \ln \left(1 + \frac{\cos^2 t}{t}\right) dt$ est divergente.

4. Étudier la nature de $M = \int_0^{\frac{1}{2}} \frac{t+1}{t(\ln t)^2} dt$, $P = \int_0^{\infty} \frac{t^2-1}{t^{\frac{1}{2}}(t^3+1)} dt$ et $Q = \int_1^{+\infty} \left| \frac{t \sin t}{e^t} \right| dt.$



Exercise

(30pts)

Compute the following integrals:

1. $\int 1 \cdot \ln|x| dx = x \ln|x| - \int dx$

(5) $= x \ln|x| - x + C.$

by part:

(2) $\frac{1}{2}$

$v = \ln|x|$

$u' = 1$

$v' = \frac{1}{x}$

$u = x$

2. $\int (\arcsin x)^2 dx = \int t^2 \cos t dt$

$= t^2 \sin t + 2t \cos t - 2 \sin t + C$

$=$ (replace t by $\arcsin x$).

By substitution

$\arcsin x = t \Rightarrow x = \sin t$

$\Rightarrow dx = \cos t dt$

By part

t^2 $\cos t$

$2t$ $\sin t$

2 $-\cos t$

0 $-\sin t$

3. $\int \sin^2 x \cos^2 x dx = \int \left[\frac{1}{8} - \frac{1}{8} \cos 4x \right] dx$

$= \frac{1}{8} x - \frac{1}{8} \cdot \frac{1}{4} \sin(4x) + C$

By linearization:

$\sin^2 x \cos^2 x = (\sin x \cos x)^2$

$= \frac{1}{4} \sin^2(2x)$

$= \frac{1}{4} \left[\frac{1 - \cos 4x}{2} \right]$

$= \frac{1}{8} - \frac{1}{8} \cos 4x$

4. $I = \int \frac{e^x \sin x}{u' \cdot v} dx$ (two times by part).

(3) $= e^x \sin x - \int \frac{e^x \cos x}{u' \cdot v} dx$

(5) $= e^x \sin x - [e^x \cos x - \int e^x (-\sin x) dx]$

(1) $I = e^x [\sin x - \cos x] = I$

$\Rightarrow 2I = e^x [\sin x - \cos x]$

(1/2) $\Rightarrow I = \frac{e^x [\sin x - \cos x]}{2} + C.$

Exercice 11.

(10pts)

Calculer :

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k}{n \sqrt{n^2 + k^2}} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{k}{\sqrt{n^2 (1 + (k/n)^2)}}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{k/n}{\sqrt{1 + (k/n)^2}} \quad (1)$$

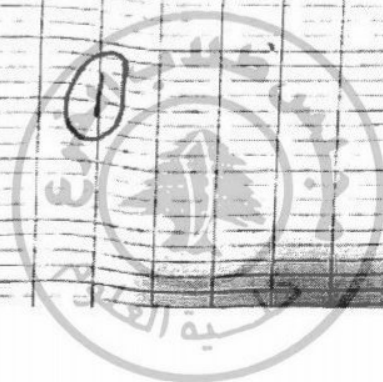
$f(x) = \frac{x}{\sqrt{1+x^2}}$ is continuous on $[0,1]$ (Quotient of continuous functions with denominator never vanishes)
Hence integrable on $[0,1]$. (2)

$$= \int_0^1 \frac{x}{\sqrt{1+x^2}} dx \quad (1)$$

$$= \frac{1}{2} \int_0^1 \frac{d(1+x^2)}{\sqrt{1+x^2}} \quad (2)$$

$$= \frac{1}{2} \left[\frac{\sqrt{1+x^2}}{1/2} \right]_0^1 \quad (1)$$

$$= \sqrt{2} - 1.$$



iii. Let $f: [0,1] \rightarrow \mathbb{R}$ be a cont. fcn.

5)

1. S.T $\lim_{n \rightarrow \infty} \int_0^{1-\frac{1}{\sqrt{n}}} f(t^n) dt = f(0)$

$$\lim_{n \rightarrow \infty} \int_0^{1-\frac{1}{\sqrt{n}}} 1 \cdot f(t^n) dt$$

$$= \lim_{n \rightarrow \infty} (f(c^n) [1 - \frac{1}{\sqrt{n}}])$$

$$= f(0) \cdot 1 \quad \text{①} \because f \text{ cont. at } 0 \text{ from right.}$$

By MVT:

② $t \mapsto f(t^n)$ cont. on $[0, 1 - \frac{1}{\sqrt{n}}] \subseteq [0,1]$.

③ $t \mapsto g(t) = 1$ is of const sign on $[0, 1 - \frac{1}{\sqrt{n}}]$.

$$\exists 0 \leq c \leq 1 - \frac{1}{\sqrt{n}} \text{ s.t. } \text{①}$$

$$\int_0^{1-\frac{1}{\sqrt{n}}} 1 \cdot f(t^n) dt = f(c^n) \cdot \int_0^{1-\frac{1}{\sqrt{n}}} 1 dt$$

$$= f(c^n) [1 - \frac{1}{\sqrt{n}}]$$

2. Deduce the value of

$$\lim_{n \rightarrow \infty} \int_0^1 f(t^n) dt.$$

$$\text{③} \int_0^1 f(t^n) dt = \int_0^{1-\frac{1}{\sqrt{n}}} f(t^n) dt + \int_{1-\frac{1}{\sqrt{n}}}^1 f(t^n) dt$$

$$= u_n + v_n$$

② $u_n \xrightarrow{n \rightarrow \infty} f(0)$ (part 1.)

$$|v_n| = \left| \int_{1-\frac{1}{\sqrt{n}}}^1 f(t^n) dt \right|$$

$$\text{②} \leq \int_{1-\frac{1}{\sqrt{n}}}^1 |f(t^n)| dt$$

$$\text{②} \leq M \int_{1-\frac{1}{\sqrt{n}}}^1 1 dt = M [1 - \frac{1}{\sqrt{n}}]$$

Hence $v_n \rightarrow 0$ as $n \rightarrow \infty$.

Thus $\int_0^1 f(t^n) dt \xrightarrow{n \rightarrow \infty} f(0)$

$$0 \leq c \leq 1 - \frac{1}{\sqrt{n}}$$

$\downarrow$

$$\text{①} \quad 0 \leq c^n \leq (1 - \frac{1}{\sqrt{n}})^n$$

$\Downarrow$

$$\lim_{n \rightarrow \infty} c^n = 0 \quad \left(\because (1 - \frac{1}{\sqrt{n}})^n \xrightarrow{n \rightarrow \infty} e^{-\sqrt{n}} \rightarrow 0 \right)$$

(f cont. on $[0,1] \Rightarrow f$ bdd on $[0,1]$)
 $\Rightarrow \exists M > 0 \text{ s.t. } |f(t)| \leq M \quad \forall t \in [0,1].$

exercice III.
10 pts

1. $I = \int_1^{+\infty} \frac{\cos t}{t} dt$ is convergent by ^① Abel Dirichlet test

• $g(t) = \frac{1}{t} \downarrow \& \xrightarrow{t \rightarrow +\infty} 0$

① + ①

• $F(t) = \int_1^t \cos t dt$ is bounded on $[1, +\infty[$

② because $|F(t)| = |\sin t - \sin 1| \leq |\sin t| + |\sin 1| \leq 1 + 1 \leq 2$

• $f(t) = \cos t$ & $g(t) = \frac{1}{t}$ are cont, hence
the int. on $[1, +\infty[$

$$J = \int_1^{+\infty} \frac{\cos^2 t}{t^2} dt$$

$$0 \leq \frac{\cos^2 t}{t^2} \leq \frac{1}{t^2}$$

& both dec. on $[1, +\infty[$

Since ^② $\int_1^{+\infty} \frac{1}{t^2}$ is convergent (By Riemann with $\alpha = 2 > 1$)

then ^② J is convergent.

2. $K = \int_1^{+\infty} \ln(1 + \frac{\cos t}{t}) dt$; $f(t) = \ln(1 + \frac{\cos t}{t})$, $f \in C[1, +\infty[$

$\Rightarrow f \in R_{loc}[1, +\infty[$

② $f(t) = \ln(1 + \frac{\cos t}{t}) = \frac{\cos t}{t} - \frac{\cos^2 t}{2t^2} + o(\frac{\cos t}{t})$

Since,

$I = \int_1^{+\infty} \frac{\cos t}{t}$ conv.

$\frac{1}{2} J = \int_1^{+\infty} \frac{\cos^2 t}{2t^2} dt$ conv } ④

and $\int_1^{+\infty} \frac{\cos^2 t}{t^2} dt$ is absol conv $\Rightarrow \int_1^{+\infty} o(\frac{\cos t}{t}) dt$ is absol conv

Then K is convergent ①